SI Units

Reference: Applied Mechanics: Hannah & Hillier and Reed's Vol. 2

The course adopts the **Système International (SI)** of units. The base units most relevant to mechanics are:

Quantity	Unit	Symbol
Mass	kilogram	kg
Length	metre	m
Time	second	s
Force	newton	N
Energy / Work	joule	J
Power	watt	W
Pressure / Stress	pascal	Pa

Derived units follow from these. For example:

$$1\ \mathrm{N} = 1\ \mathrm{kg}\cdot\mathrm{m/s}^2 \qquad 1\ \mathrm{Pa} = 1\ \mathrm{N/m}^2 \qquad 1\ \mathrm{J} = 1\ \mathrm{N}\cdot\mathrm{m}$$

Scalar and Vector Quantities

Scalar quantities have magnitude only and obey ordinary arithmetic:

- Examples: mass, speed, temperature

Vector quantities have both magnitude and direction and must be combined using vector algebra:

- Examples: force, velocity, acceleration, momentum

A vector is represented graphically as an arrow — its length denotes magnitude, its orientation denotes direction.

Vector Addition

Triangle Rule

Two vectors $\mathbf{A}$ and $\mathbf{B}$ are added tip-to-tail; the resultant $\mathbf{R}$ closes the triangle:

$$\mathbf{R} = \mathbf{A} + \mathbf{B}$$

Parallelogram Rule

Both vectors are drawn from the same point; the resultant is the diagonal of the parallelogram they form.

Polygon Rule

For more than two vectors, they are placed tip-to-tail in sequence; the resultant closes the polygon from the tail of the first to the tip of the last.

Vector Subtraction

Subtracting $\mathbf{B}$ from $\mathbf{A}$ is equivalent to adding the negative of $\mathbf{B}$:

$$\mathbf{R} = \mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$$

Resolution of Vectors

Any vector $\mathbf{F}$ can be resolved into two perpendicular components. For a vector at angle θ to the horizontal:

$$F_x = F\cos\theta$$
$$F_y = F\sin\theta$$

This is the most widely used technique in the book — resolving all forces into horizontal and vertical components before applying equilibrium conditions.

Resultant of Several Vectors

To find the resultant of a system of vectors analytically:

1. Resolve each vector into x and y components
2. Sum all components in each direction:

$$\Sigma F_x = F_{1x} + F_{2x} + \cdots \qquad \Sigma F_y = F_{1y} + F_{2y} + \cdots$$

3. Find the magnitude of the resultant:

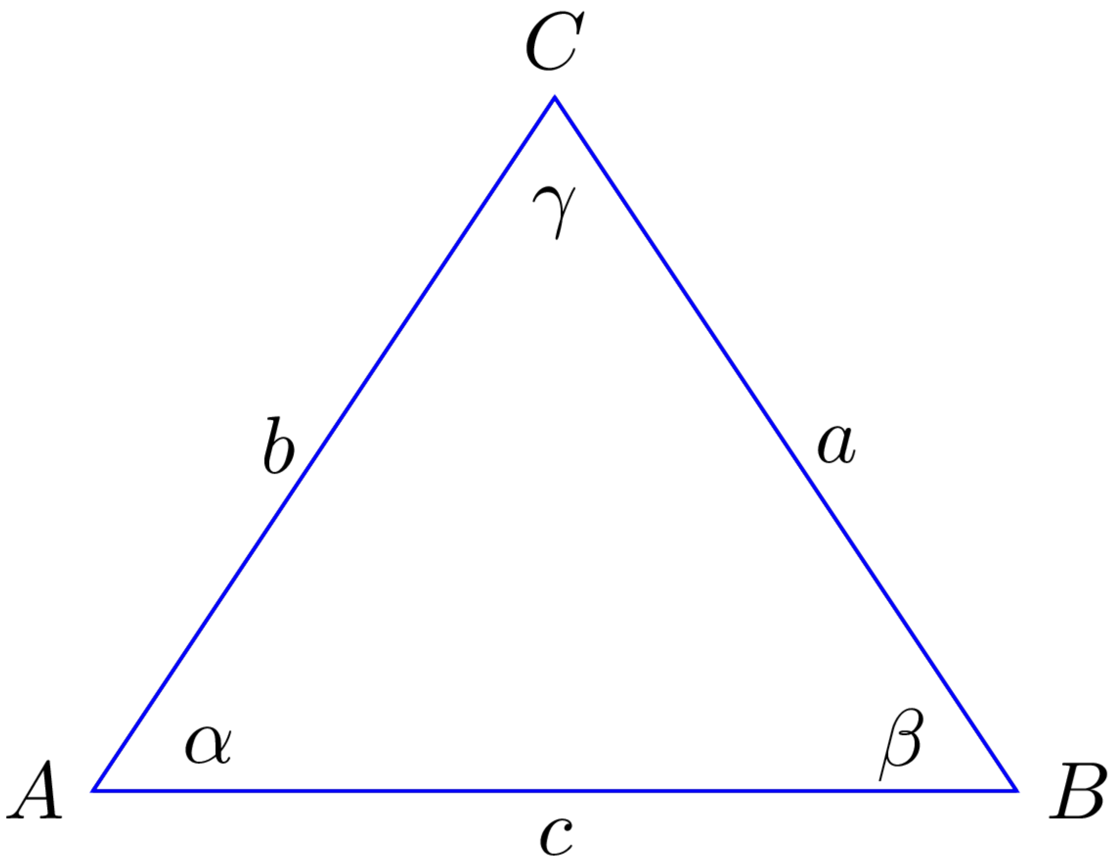
$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$$

4. Find the direction:

$$\theta = \arctan\left(\frac{\Sigma F_y}{\Sigma F_x}\right)$$

The Laws of Sines and Cosines

For cases involving triangles of vectors, these laws provide an analytical alternative to graphical methods.



Sine rule:

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$$

Cosine rule:

$$c^2 = a^2 + b^2 - 2ab \cos \gamma$$

Summary of Equations

Concept	Equation
Horizontal component	$F_x = F \cos \theta$
Vertical component	$F_y = F \sin \theta$
Resultant magnitude	$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$
Resultant direction	$\theta = \arctan(\Sigma F_y / \Sigma F_x)$
Cosine rule	$c^2 = a^2 + b^2 - 2ab \cos C$